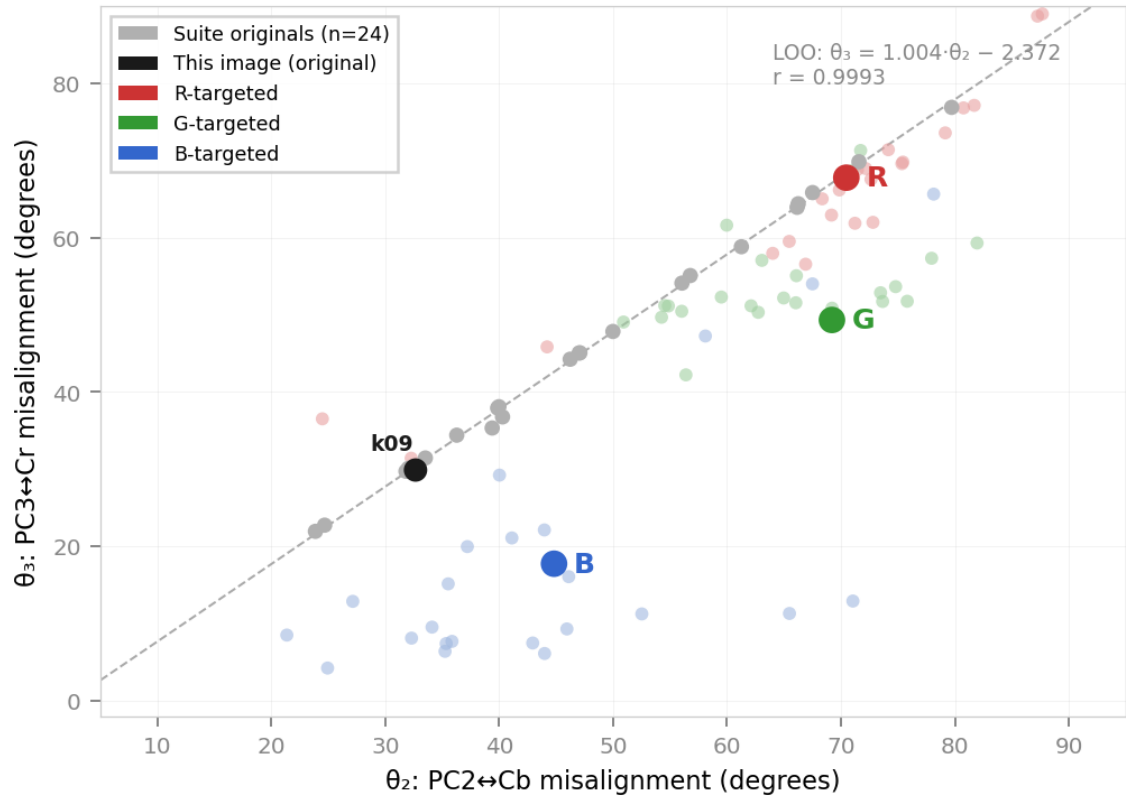


Compression Profile: KODIM09512x768 | 24-bit RGB | PNG (lossless)

sailboats under spinnakers | Weakly One-Dimensional (PC1 = 90.85%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9486	0.7330	0.8560	0.846	32.65°	29.89°	-0.52°
R-targeted	0.0268	-0.0063	0.9429	0.325	70.47°	67.8°	-0.58°
G-targeted	-0.2015	0.9665	-0.2037	0.457	69.21°	49.35°	-17.76°
B-targeted	0.9321	-0.2114	-0.0638	0.402	44.81°	17.74°	-24.88°

R-targeted isolates red (R-G, R-B < 0.03) while preserving G-B coupling at 0.943.
G-targeted isolates green (R-G, G-B < 0.20) while preserving R-B at 0.967.
B-targeted preserves R-G at 0.932 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM09512x768 | 24-bit RGB | PNG (lossless)

sailboats under spinnakers | Weakly One-Dimensional (PC1 = 90.85%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

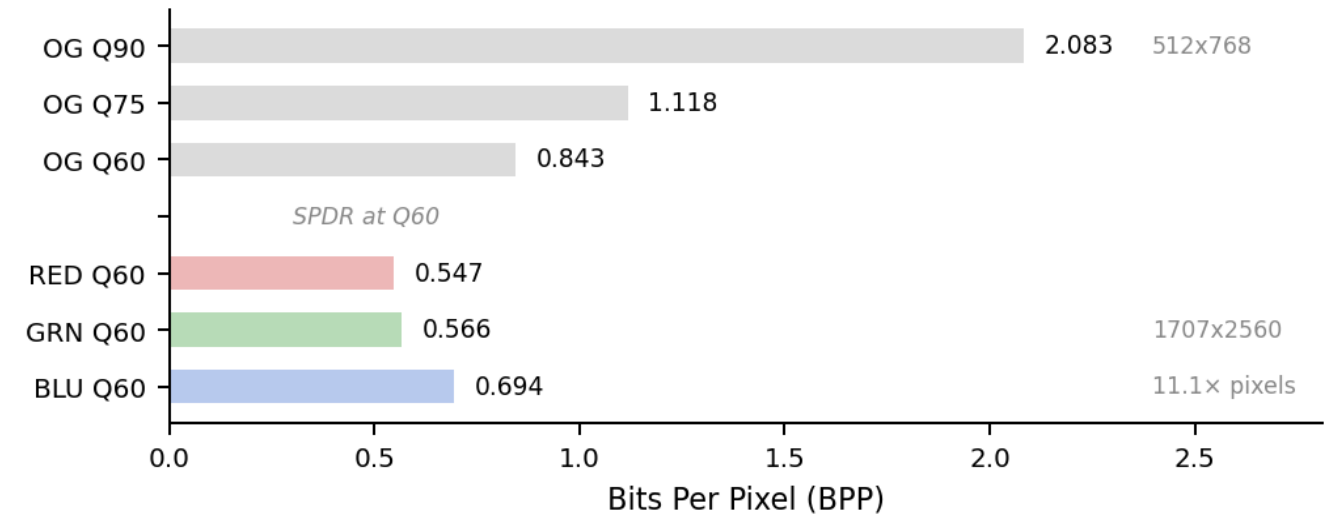
ORIGINAL (kodim09)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	512x768	569.2 KB	11.859	393,216
JPEG Q90	JPEG	512x768	100.0 KB	2.083	393,216
JPEG Q75	JPEG	512x768	53.7 KB	1.118	393,216
JPEG Q60	JPEG	512x768	40.5 KB	0.843	393,216

REDISTRIBUTED (kodim09 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	1707x2560	291.9 KB	0.547	4,369,920
GRN Q60	JPEG	1707x2560	301.8 KB	0.566	4,369,920
BLU Q60	JPEG	1707x2560	370.4 KB	0.694	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM09512x768 | 24-bit RGB | PNG (lossless)

sailboats under spinnakers | Weakly One-Dimensional (PC1 = 90.85%)

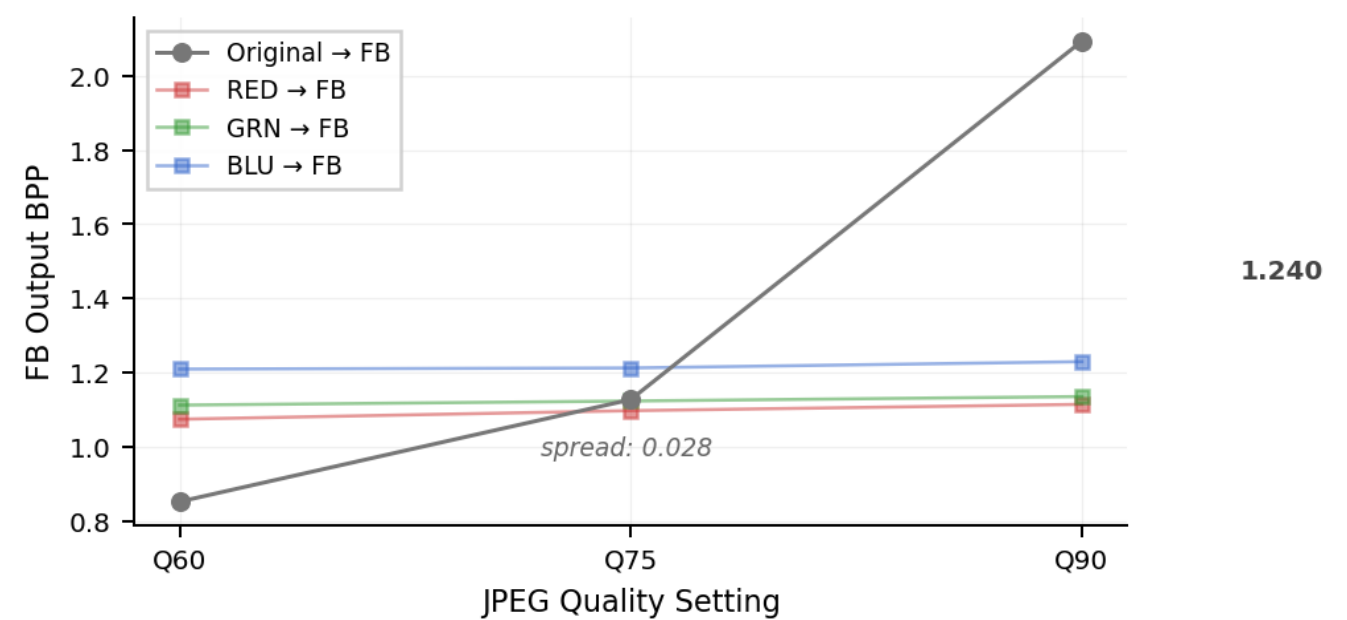
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	512x768	512x768	1.973	–
R-targeted → FB	1707x2560	1366x2048	1.115	43.5%
G-targeted → FB	1707x2560	1366x2048	1.136	42.4%
B-targeted → FB	1707x2560	1366x2048	1.230	37.7%
ABC (no perturbation) → FB	1707x2560	1366x2048	1.151	41.7%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.8440 vs 0.846 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.240 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.028 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim09

Best BPP (Q60, direct)	0.547	1707x2560, 4,369,920 px
Best BPP through FB	1.115	1366x2048 output
vs OG Q90 → FB	46.7%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216